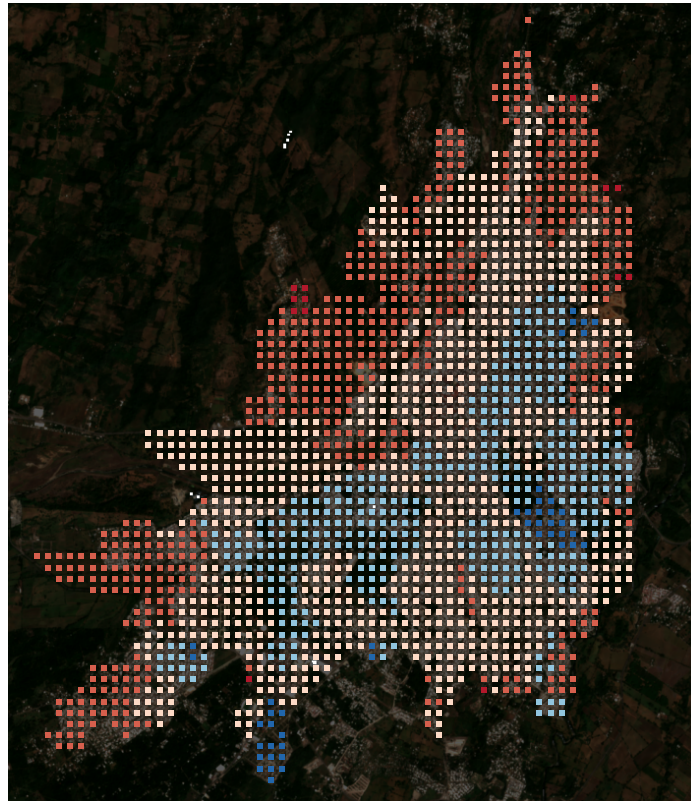


Supplementary Information

Mapping intra-urban social deprivation from Sentinel imagery using only municipal
aggregate labels

AGEB ground truth (held out)



Supplementary Figure 1: Tract-level ground truth for Tapachula, on the color scale of main-text Fig. 1c. Every token inherits the held-out GRS grade of the AGEB it falls in.

Supplementary Table 1: The supervision-efficiency curve (main-text Fig. 2a). Mean over three seeds; validation cities were used for selection, test cities were scored once with frozen models.

Bags	Validation within- ρ (s.d.)	Test within- ρ
50	0.150 (0.021)	0.156
100	0.141 (0.021)	0.158
200	0.151 (0.022)	0.161
400	0.182 (0.020)	0.192
771	0.227 (0.010)	0.185 (retrained); 0.196 (saved weights)

Supplementary Table 2: Agreement by tract size (main-text Limits). Within-municipality Spearman over municipalities with at least ten tracts in the bin.

Tokens per tract	Tracts	Median population	Municipalities	Within- ρ
1–3	567	211	15	0.257
4–9	849	1,948	16	0.338
10–19	999	2,644	20	0.352
20–49	464	3,231	14	0.432
≥ 50	79	3,561	2	0.127

Supplementary Table 3: Municipalities excluded from the expansion for insufficient optical depth or persistent acquisition errors; the guard requires eight observations per pixel and excludes shallow ground instead of compositing it.

Municipality	Reason
Amozoc; Comalcalco; Hidalgo del Parral; Lagos de Moreno; Nuevo Casas Grandes	composite below the eight-observation depth floor (optical arm)
La Independencia	inconsistent tract geometry
El Marqués	repeated acquisition timeouts